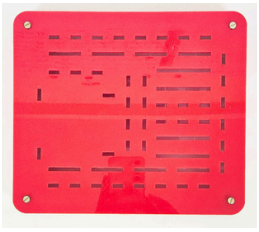
SCIENCEUTSAV®
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Session 3

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


2


3


4


5


6

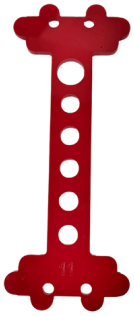

7


8


9


10


11


12


1

Spacers

2

L Clamp

3

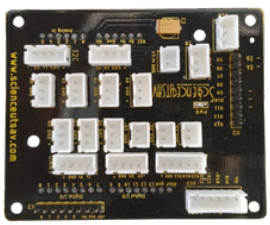
Spacer Screws

4

Nuts

5

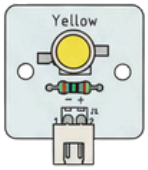
Screw

6

Arduino With Shield

7

Arduino cable

ELECTRONIC COMPONENTS



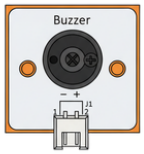
1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



RGB LED

A RGB DIP LED is a single LED with red, green, and blue lights inside; it has one common leg and three color legs, and by turning different color legs ON or OFF, we can create many colors.



BUZZER PCB

A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



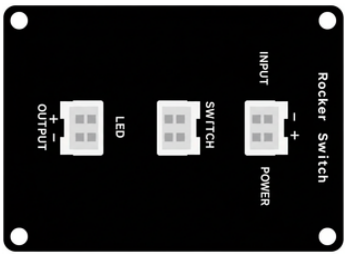
ROCKER SWITCH

A rocker switch is an electrical device that allows you to turn a circuit ON or OFF by pressing one side of the switch. It is frequently utilized in electronic projects and appliances for convenient power control.



Push Switch

A push switch is a button-style switch that operates only when pressed.
It provides temporary input in circuits, such as starting a device or triggering an action.



rocker switch PCB

A rocker switch PCB is a small circuit board designed to hold and connect a rocker switch easily in electronic projects.
It provides proper connections and makes the switch installation neat, stable, and easy to use.



2 PIN RMC

Used to connect the 1-watt LEDs and the buzzer.

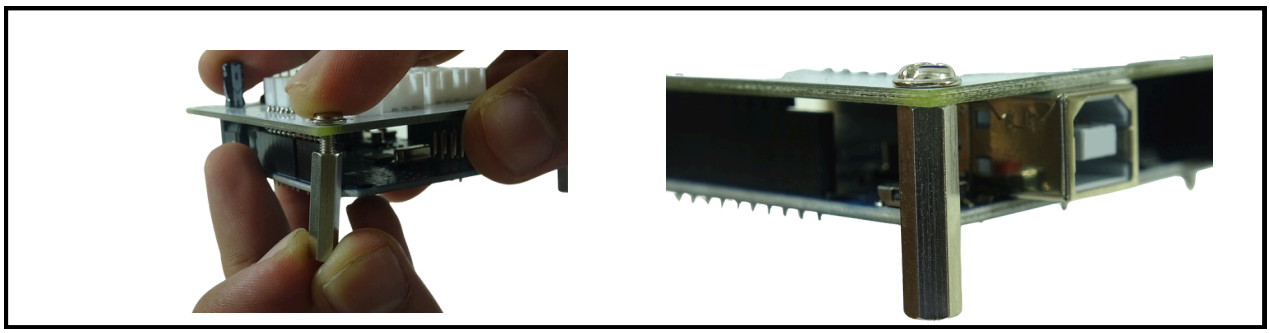


4 PIN RMC

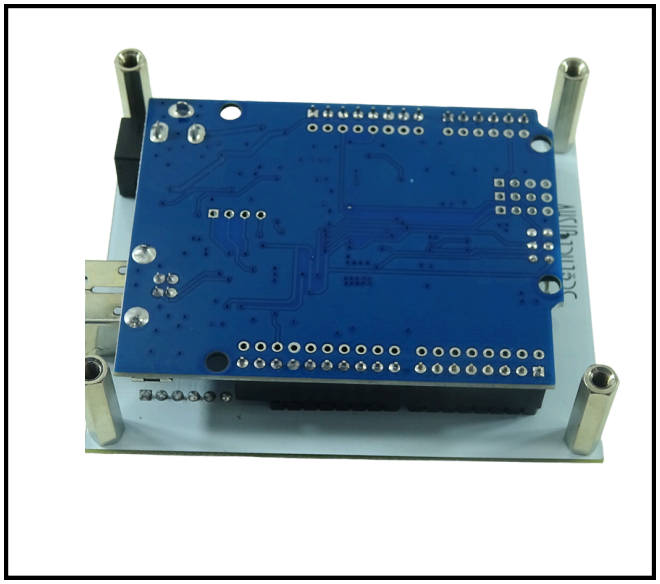
Used to connect the RGB dip led

Procedure :

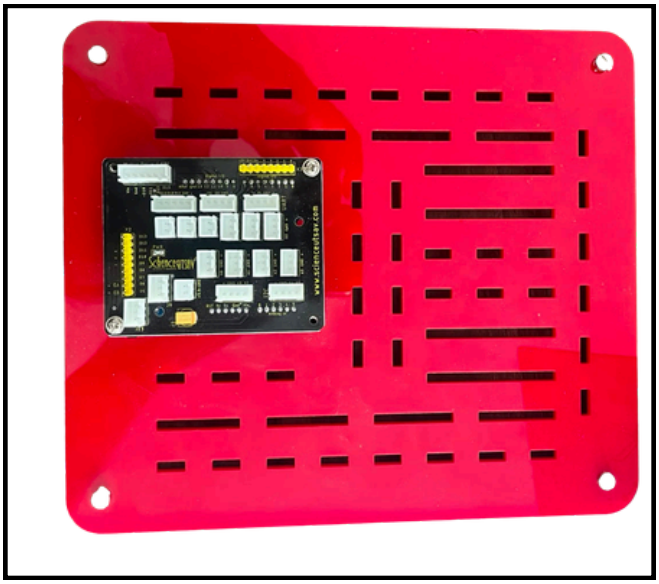
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



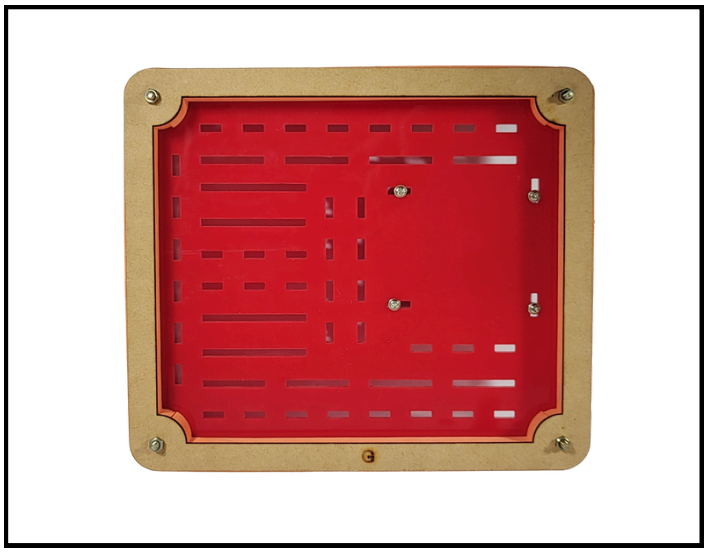
Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



Step 3 : Place the Arduino Shield to the base board as shown below

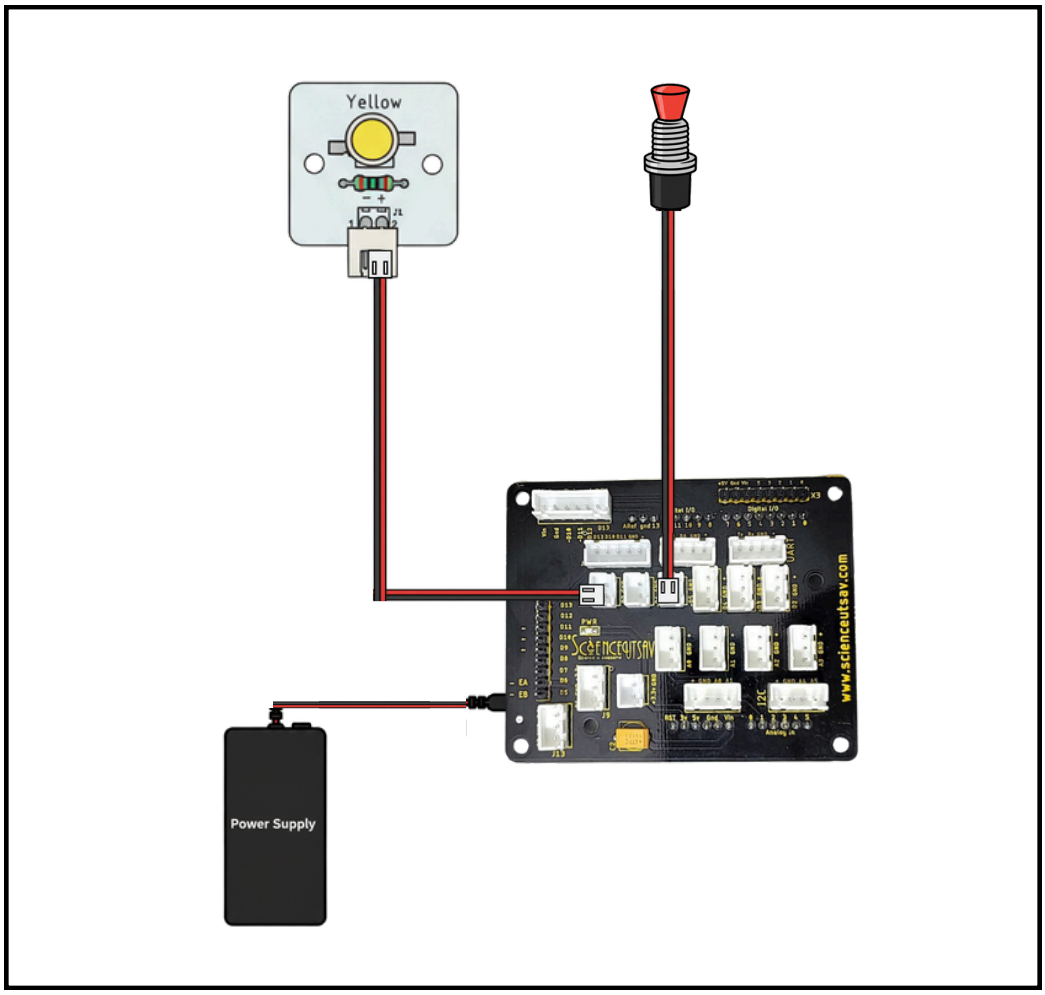


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

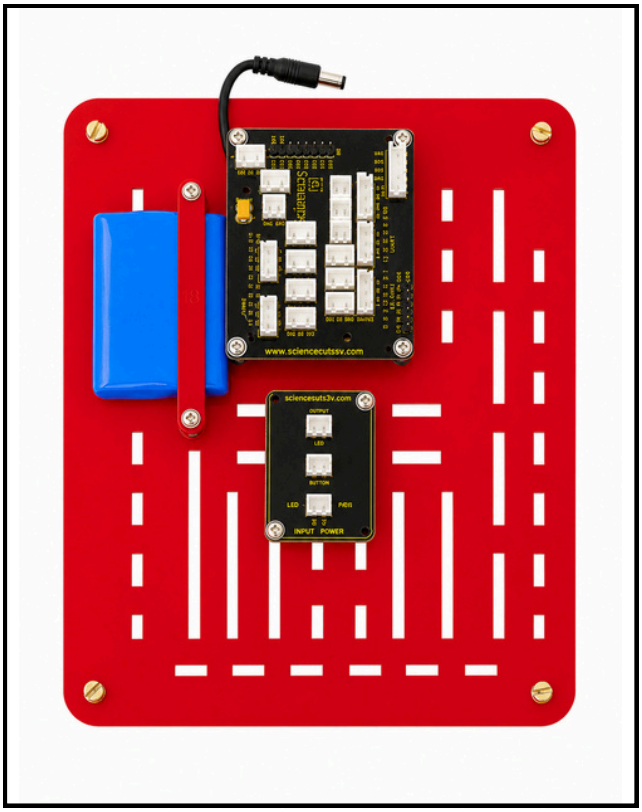
The circuit diagram of the project is given below



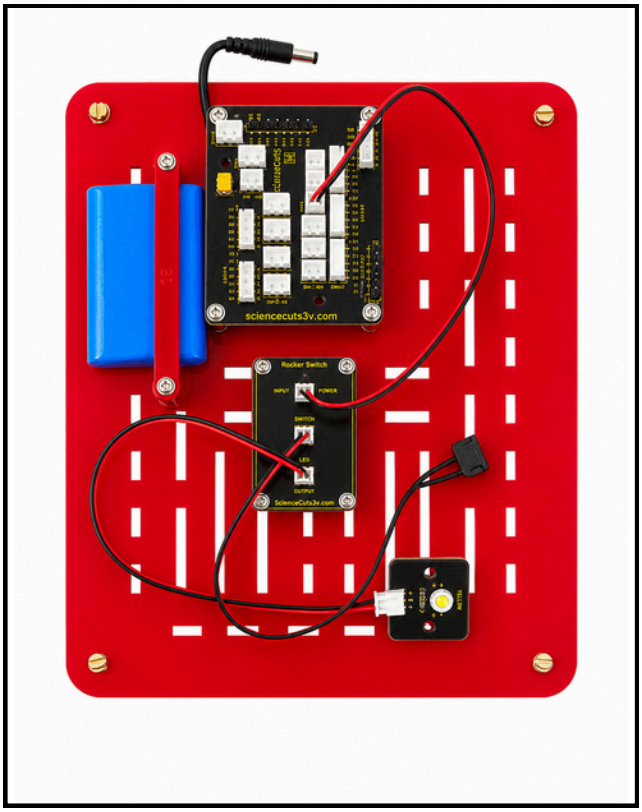
CONNECTION:

1. 1W LED IN D7
2. PUSH SWITCH IN D5

Step 1- Fix the rocker switch PCB to the board



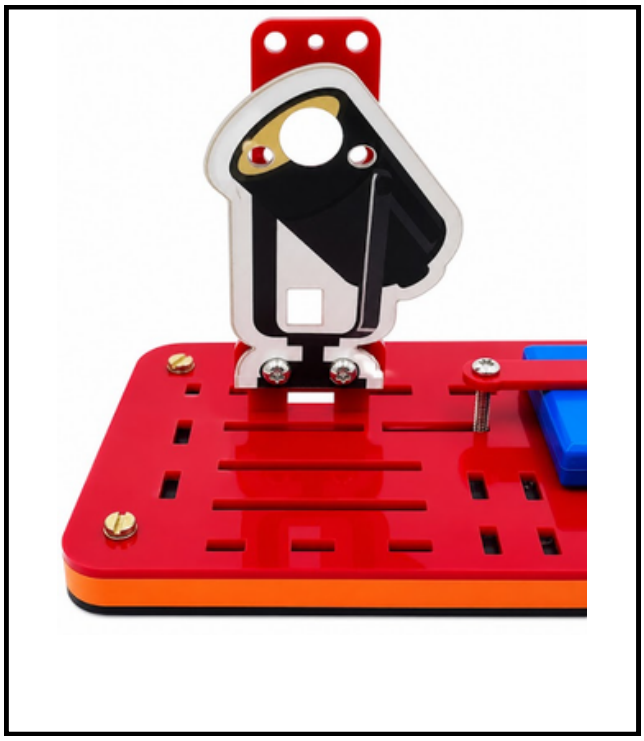
Step 2- Connect the rocker switch and the LED to the PCB



Step 3- Attach the switch holder as shown

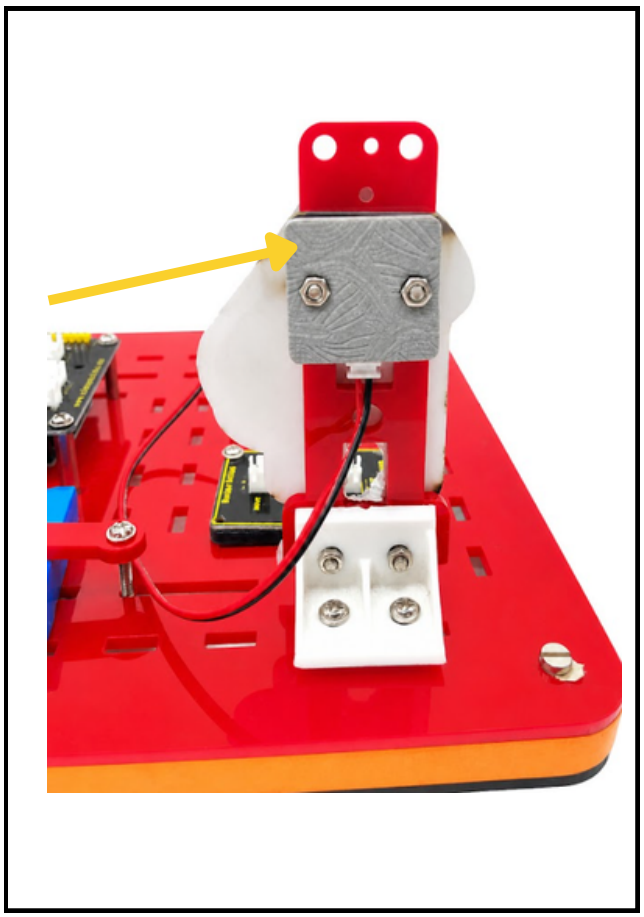


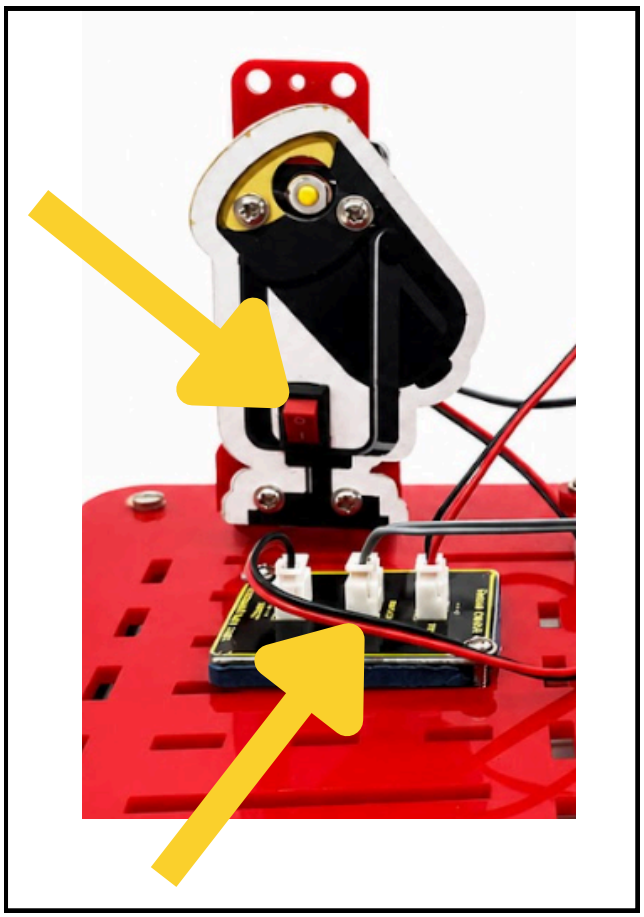
Step 4- insert 3 mm bolt in template with structure



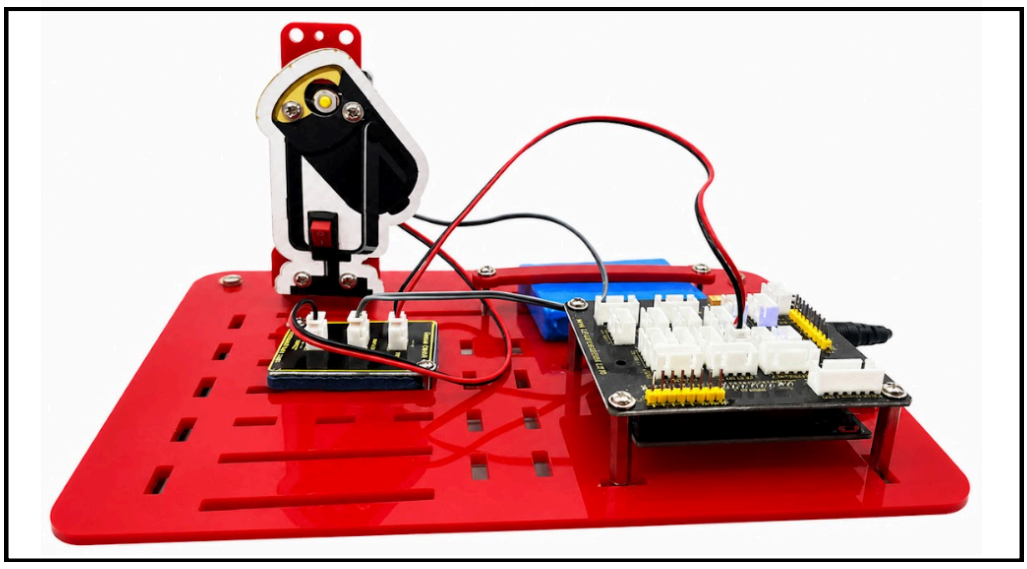
Step 5-
connect led pcb from back side of structure and attach nut

Step 6- Insert the rocker switch into the structure from the front side of the template.
Connect the other end of the rocker switch to the switch slot on the rocker switch PCB.

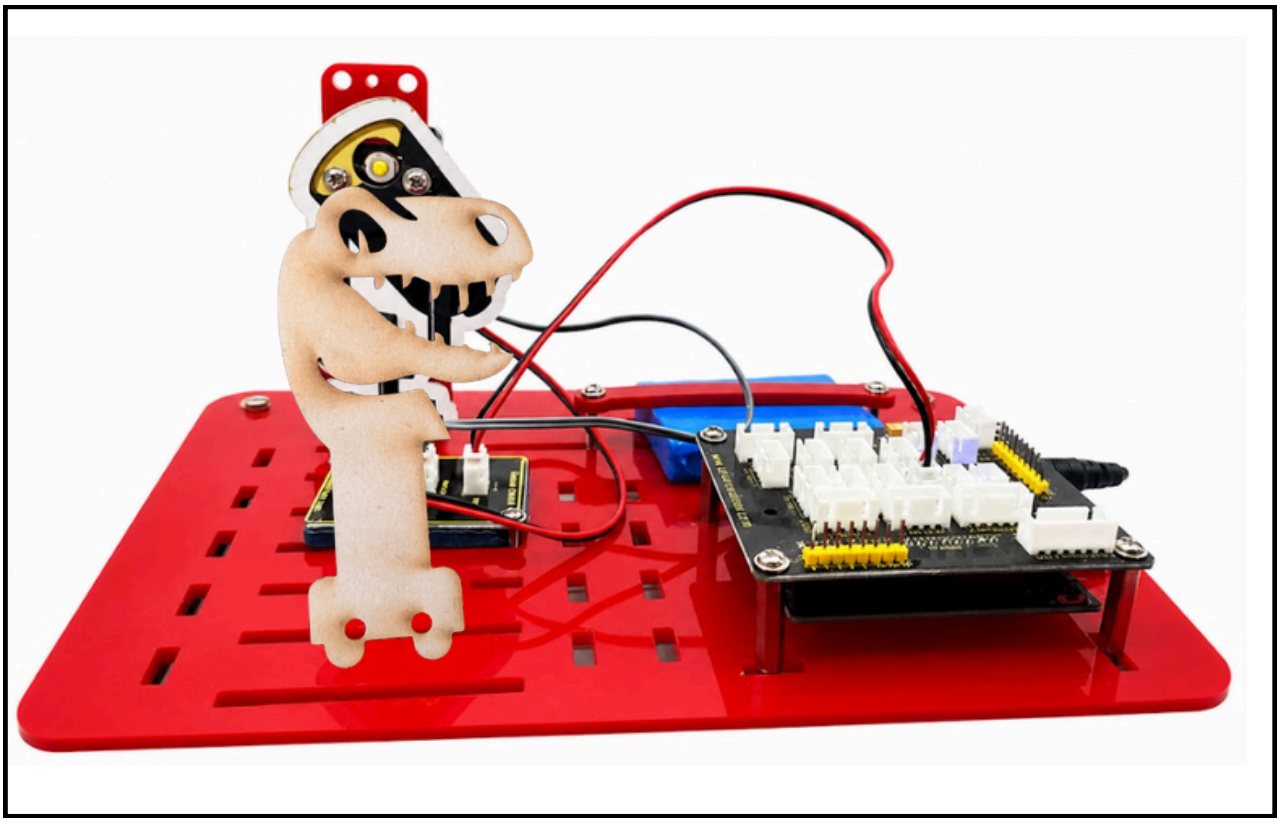


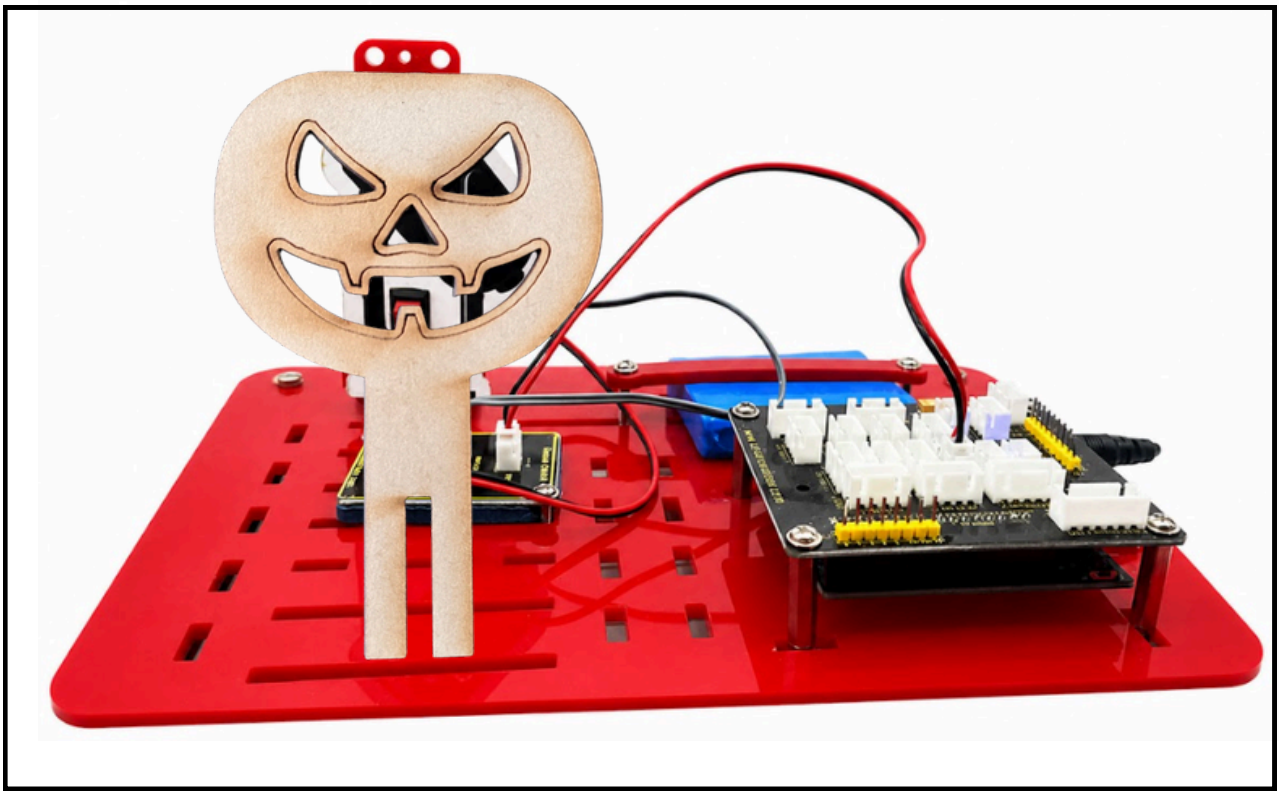


Your project is now ready for exploration!



Turn on the switch and position the dinosaur and Halloween structure in front of the light. Then, take a moment to observe the shadow that is cast.





code

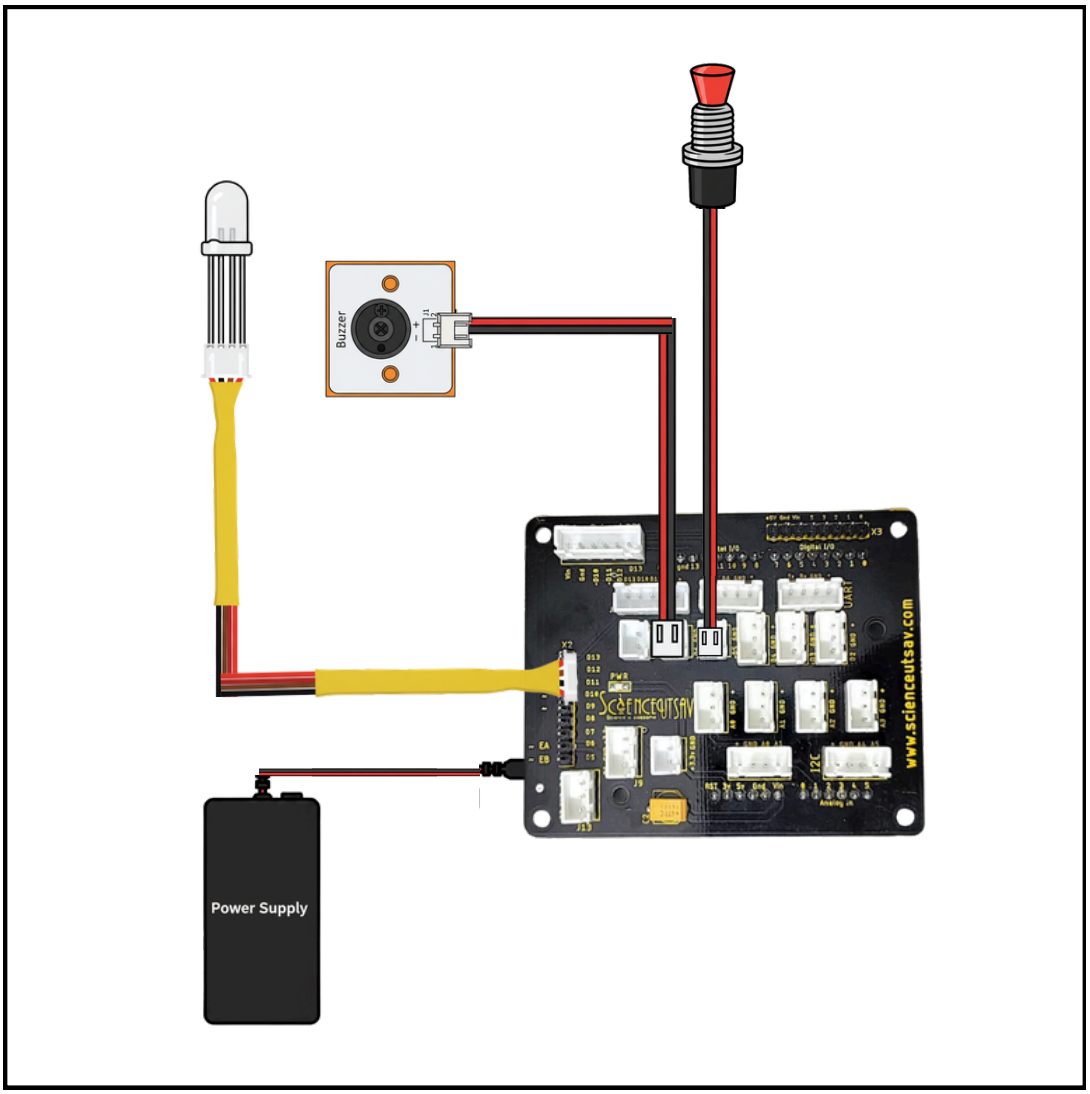
```
void setup()
{
pinMode(5,OUTPUT):

}

void loop()
{
digitalWrite(5,HIGH);
}
```

INTERMEDIATE PROJECT

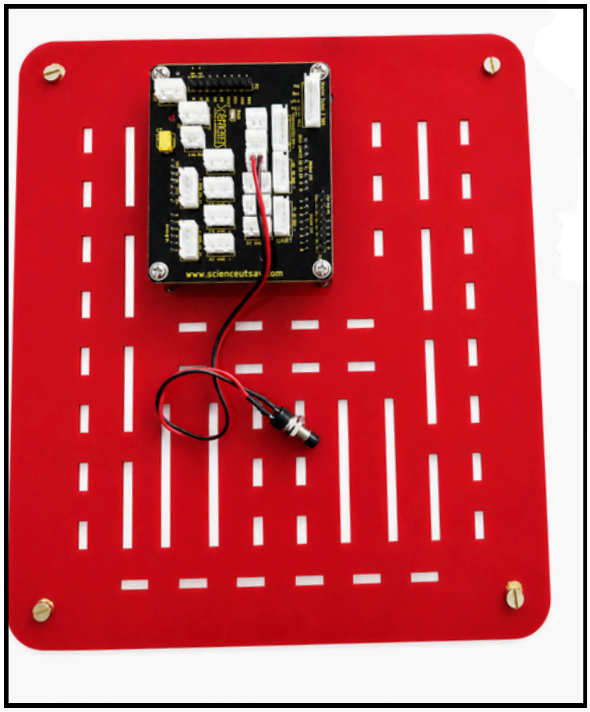
The circuit diagram of the project is given below



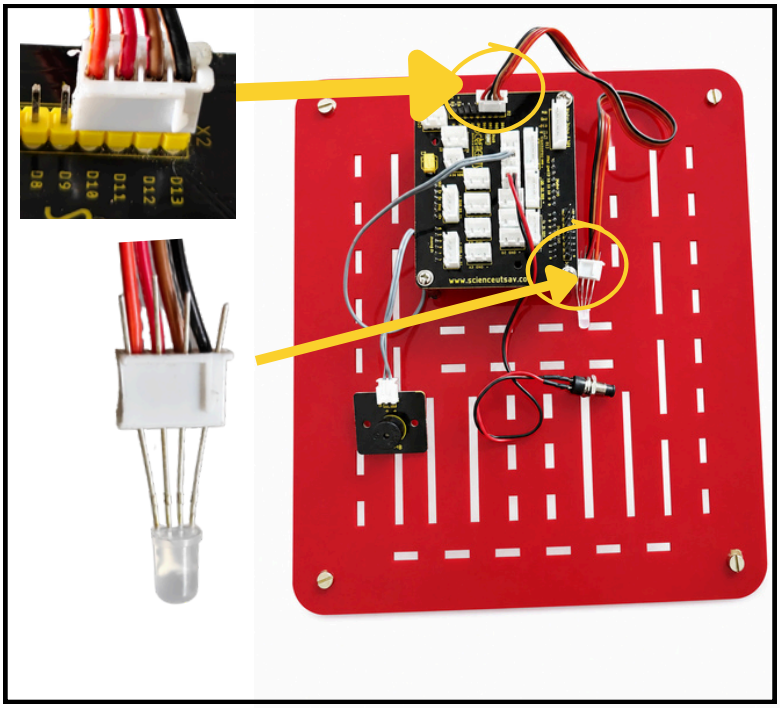
CONNECTION:

1. RGB LED IN D10, D11, D12, D13
2. PUSH SWITCH IN D5
3. BUZZER IN D6

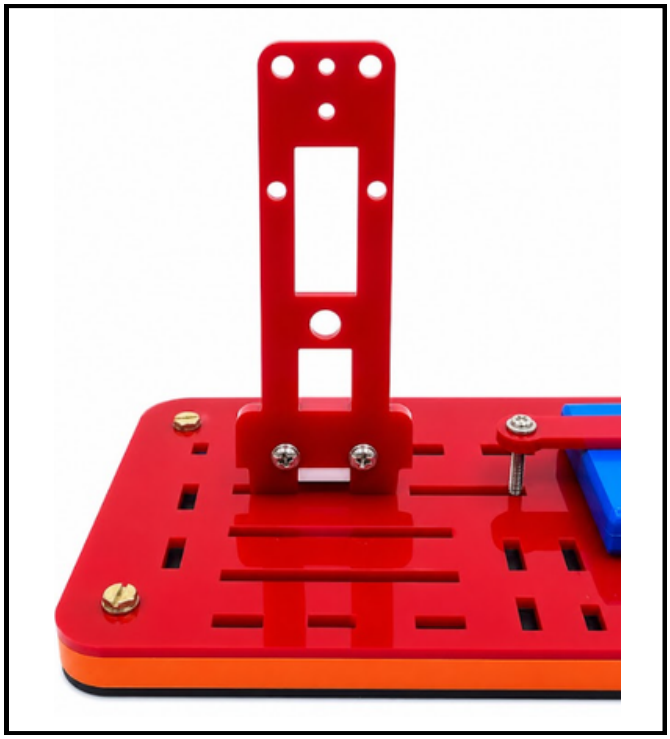
Step1- Fix the push switch PCB to the D5 of shield



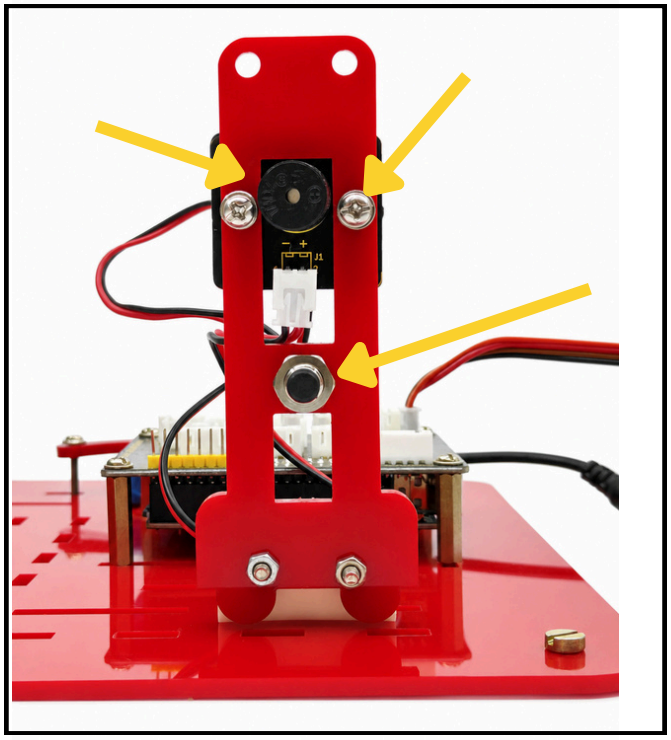
Step 2- Connect the buzzer to pin D6. Insert the RGB LED into the 4-pin connector and connect it to the 4-pin RMC. Finally, attach the other end of the RMC to the D10, D11, D12, and D13 pins located on the top side of the shield.



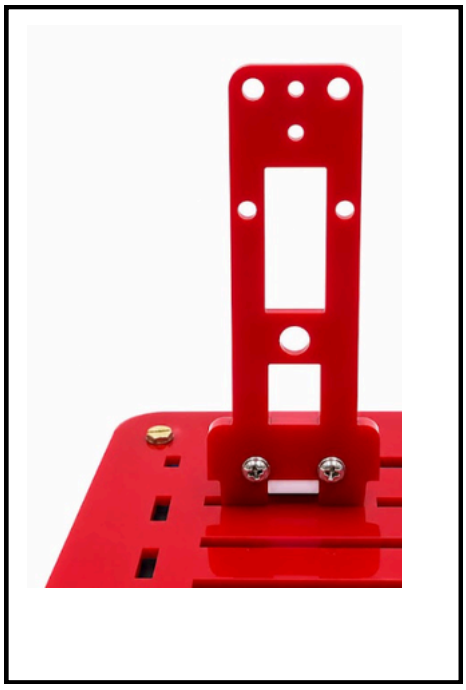
Step 3- Attach the switch holder with L clamp as shown



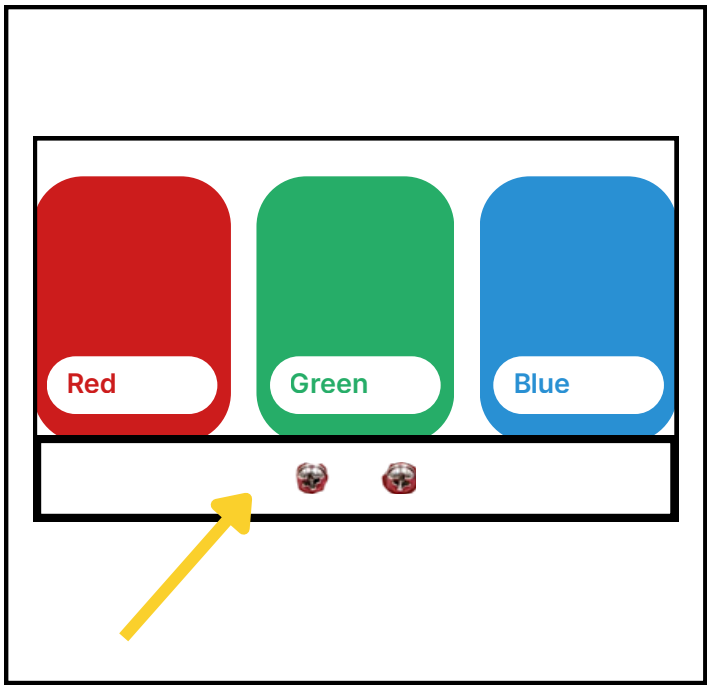
Step 4 – Secure the buzzer using a 3mm bolt and nut, then insert the push button into the same structure.



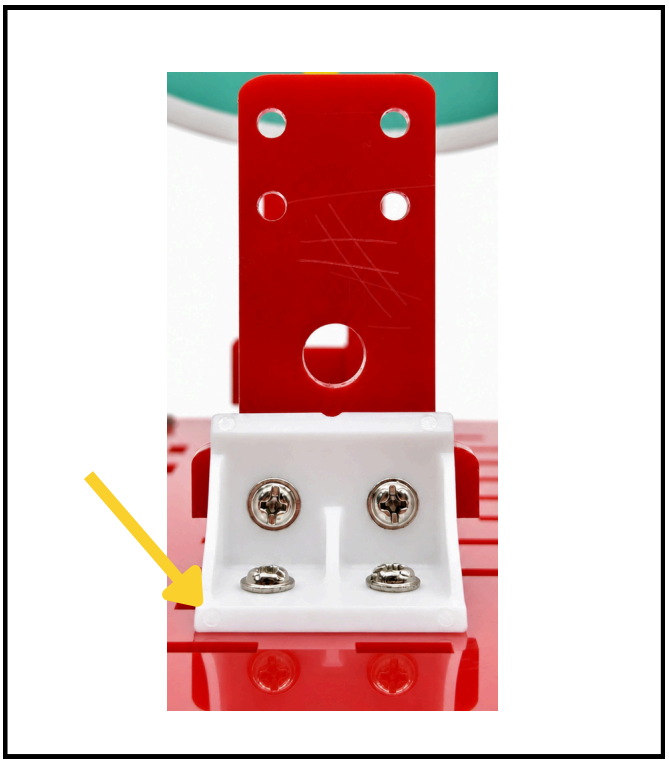
Step 5- Attach the one more switch holder with L clamp as shown



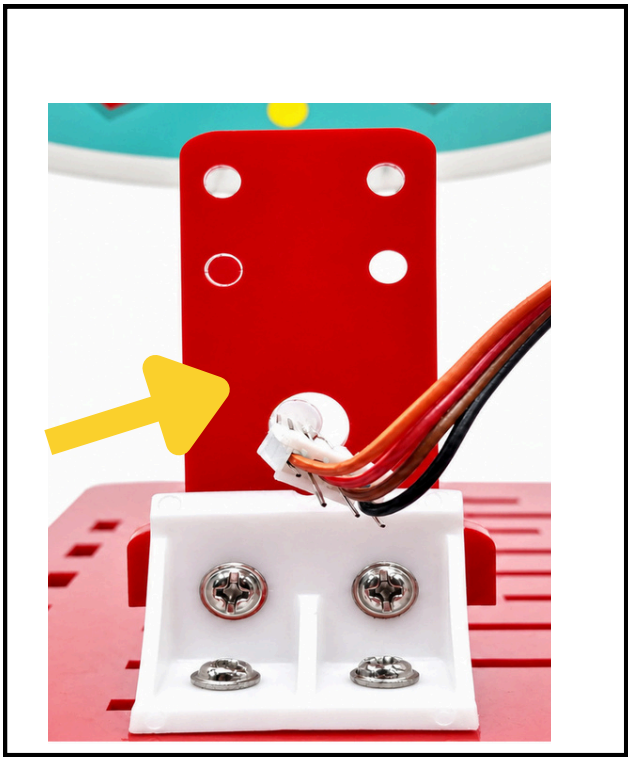
Step 6- Attach template to the structure with 3mm nut and bolt



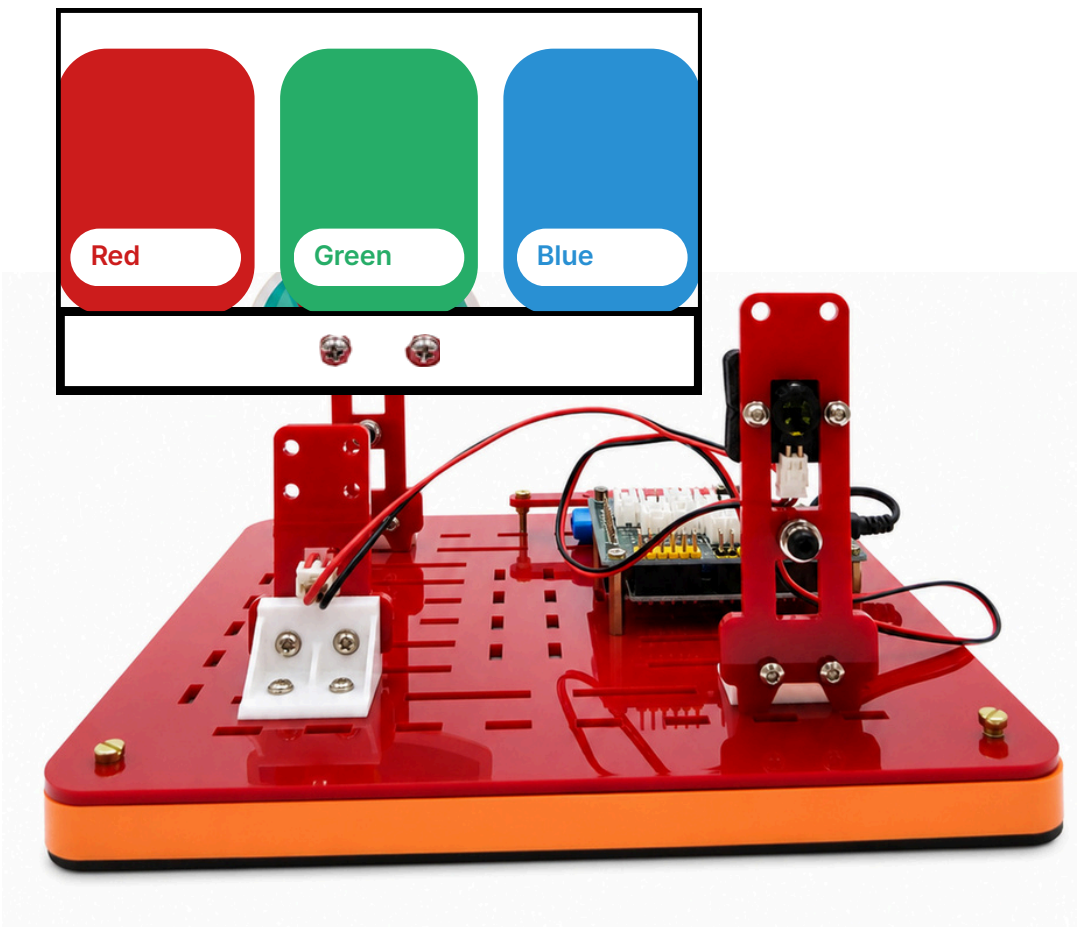
Step 7 – Attach the small stand to the board using an L clamp along with a 3mm nut and bolt.



Step 8- Insert rgb led into to the hole in the stand



Press the push switch and watch the template change through three different colors: red, green, and blue.



code

```
const int pushSwitch = 5;
const int buzzer = 6;
const int rgbRed = 10;
const int rgbGreen = 12;
const int rgbBlue = 13;

void setup() {
 pinMode(pushSwitch, INPUT_PULLUP);
 pinMode(buzzer, OUTPUT);
 pinMode(11, OUTPUT);

 pinMode(rgbRed, OUTPUT);
 pinMode(rgbGreen, OUTPUT);
 pinMode(rgbBlue, OUTPUT);

 digitalWrite(11, HIGH); // D11 HIGH forever
}

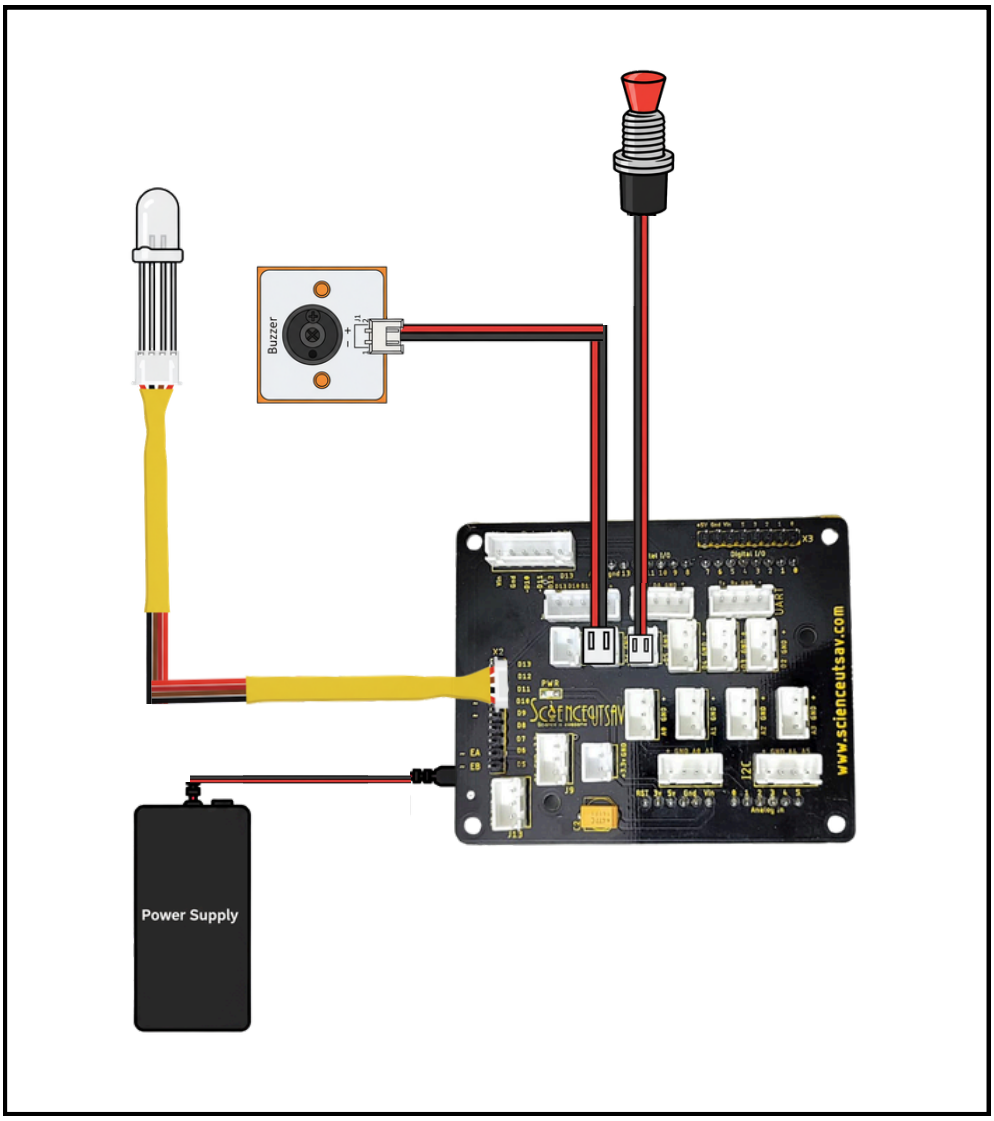
void loop() {
 int switchState = digitalRead(pushSwitch);
 if (switchState == LOW) {
 // Switch pressed
 // Red + buzzer ON
 digitalWrite(rgbRed, HIGH);
 digitalWrite(rgbGreen, LOW);
 digitalWrite(rgbBlue, LOW);
 digitalWrite(buzzer, HIGH);
 delay(500);
}
```

```
// Green + buzzer OFF
digitalWrite(rgbRed, LOW);
digitalWrite(rgbGreen, HIGH);
digitalWrite(rgbBlue, LOW);
digitalWrite(buzzer, LOW);
delay(500);

// Blue + buzzer ON
digitalWrite(rgbRed, LOW);
digitalWrite(rgbGreen, LOW);
digitalWrite(rgbBlue, HIGH);
digitalWrite(buzzer, HIGH);
delay(500);
}
else {
// Switch not pressed
digitalWrite(rgbRed, LOW);
digitalWrite(rgbGreen, LOW);
digitalWrite(rgbBlue, LOW);
digitalWrite(buzzer, LOW);
}
```

ADVANCE PROJECT

The circuit diagram of the project is given below

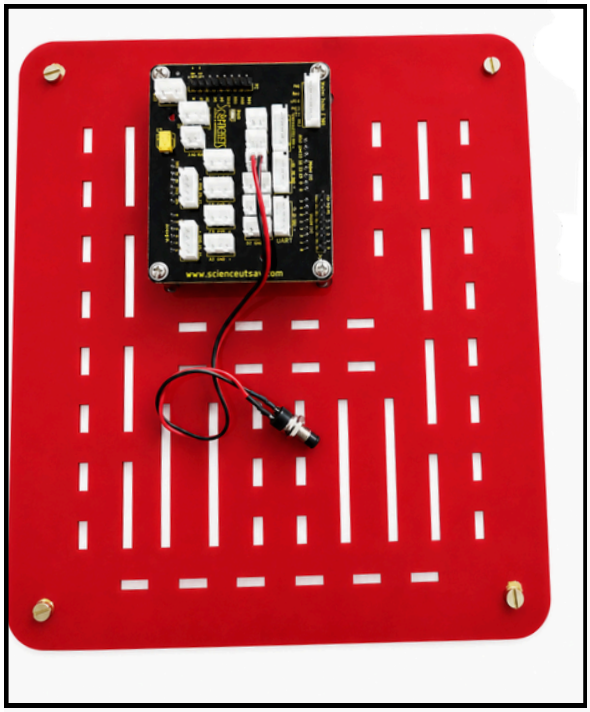


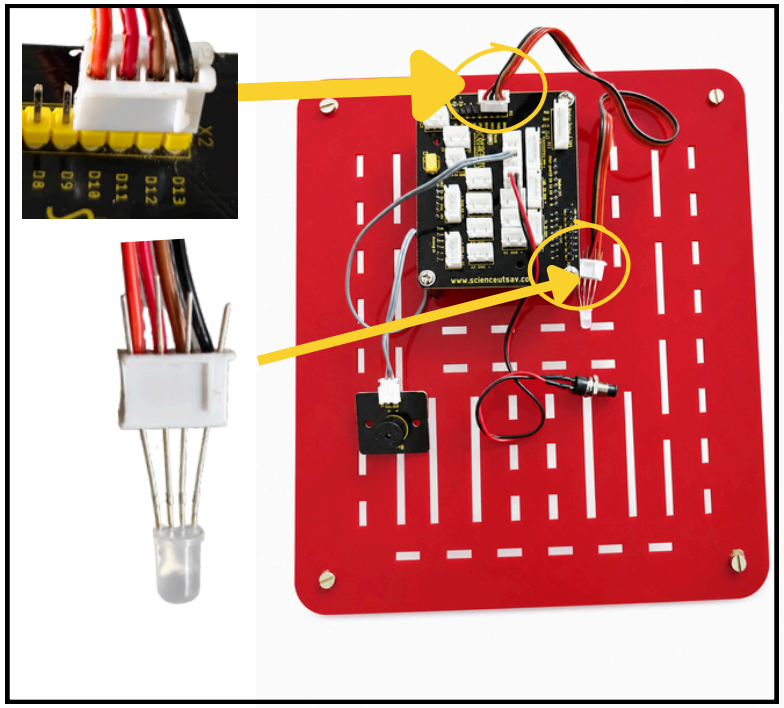
CONNECTION:

1. RGB LED IN D10, D11, D12, D13
2. PUSH SWITCH IN D5
3. BUZZER IN D6

Step1- Fix the push switch PCB to the D5 of shield

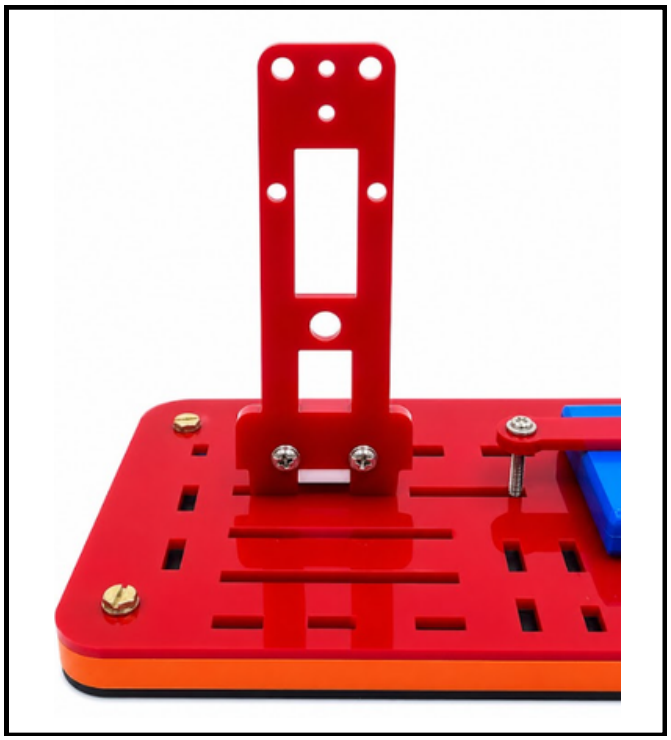
Step 2- Connect the buzzer to pin D6. Insert the RGB LED into the 4-pin connector and connect it to the 4-pin RMC. Finally, attach the other end of the RMC to the D10, D11, D12, and D13 pins located on the top side of the shield.

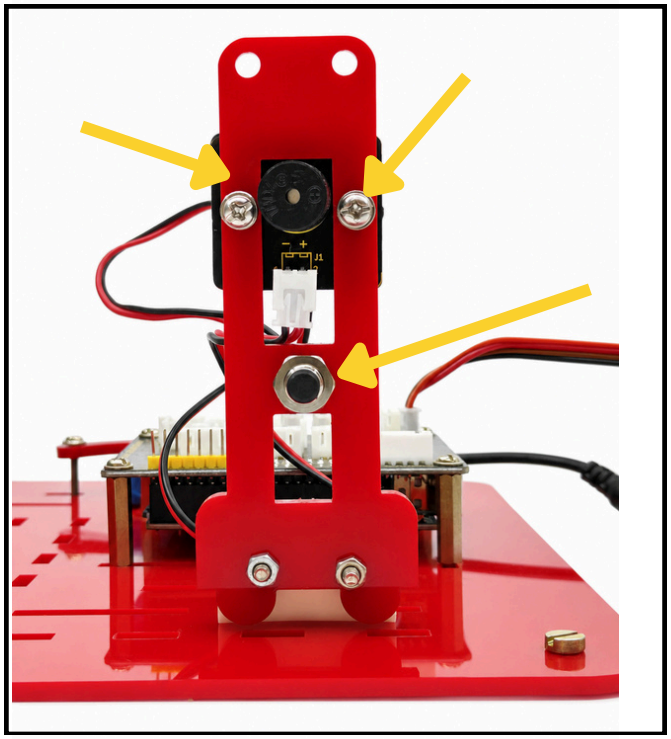




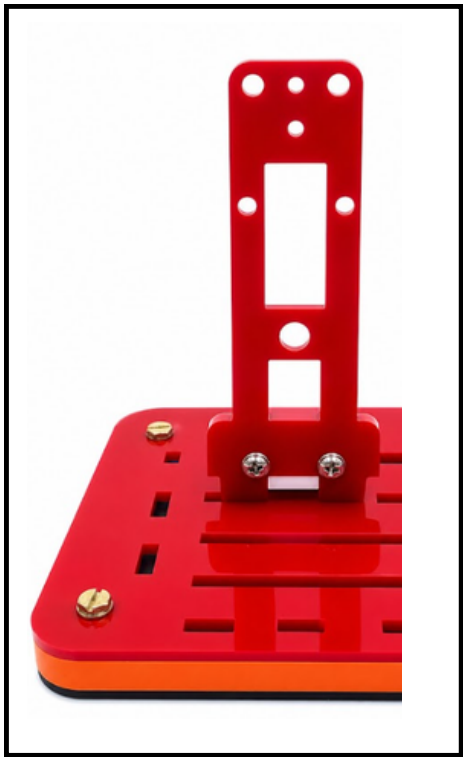
Step 3- Attach the switch holder with L clamp as shown

Step 4 – Secure the buzzer using a 3mm bolt and nut, then insert the push button into the same structure.



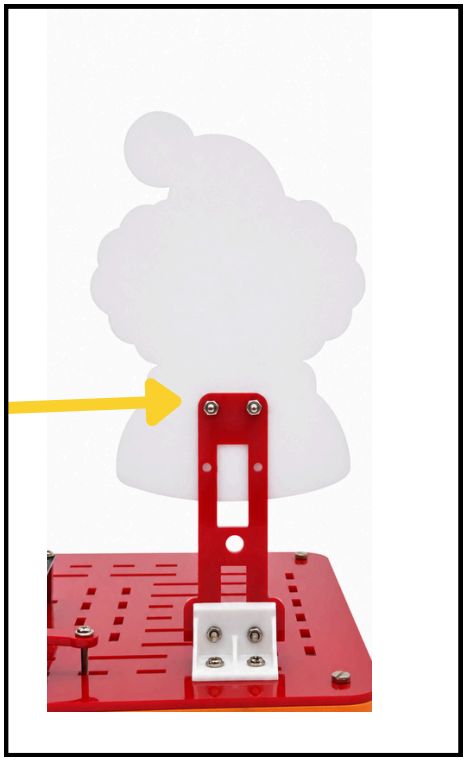


Step 5 Attach the one more switch holder with L clamp as shown

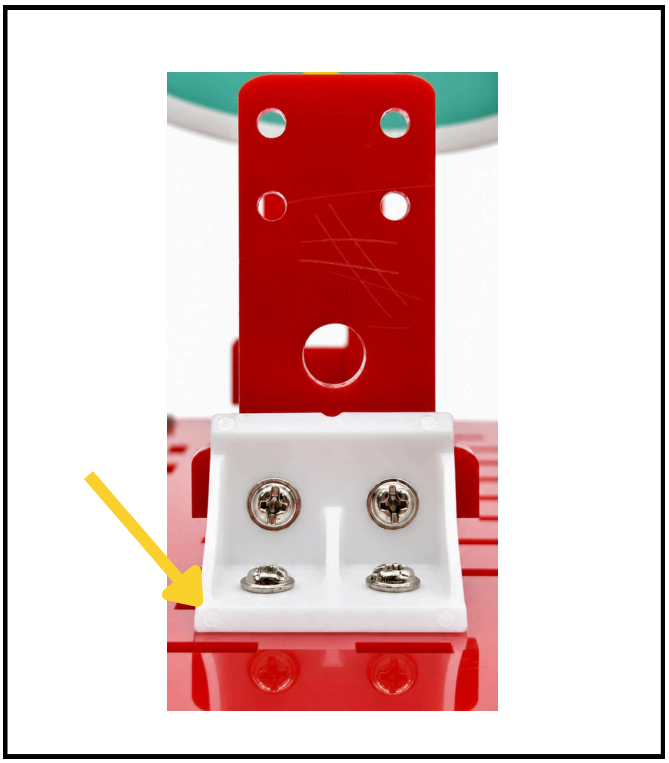


Step 6- Attach template to the structure with 3mm nut and bolt

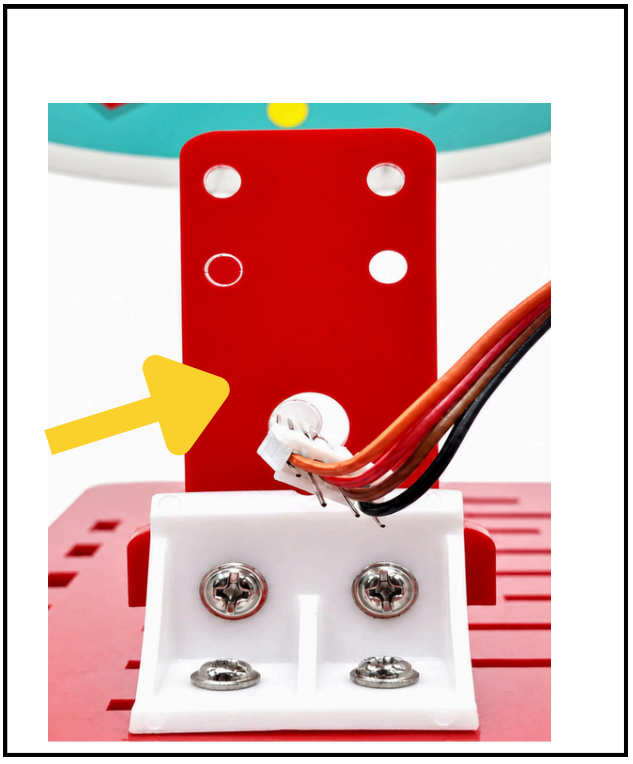




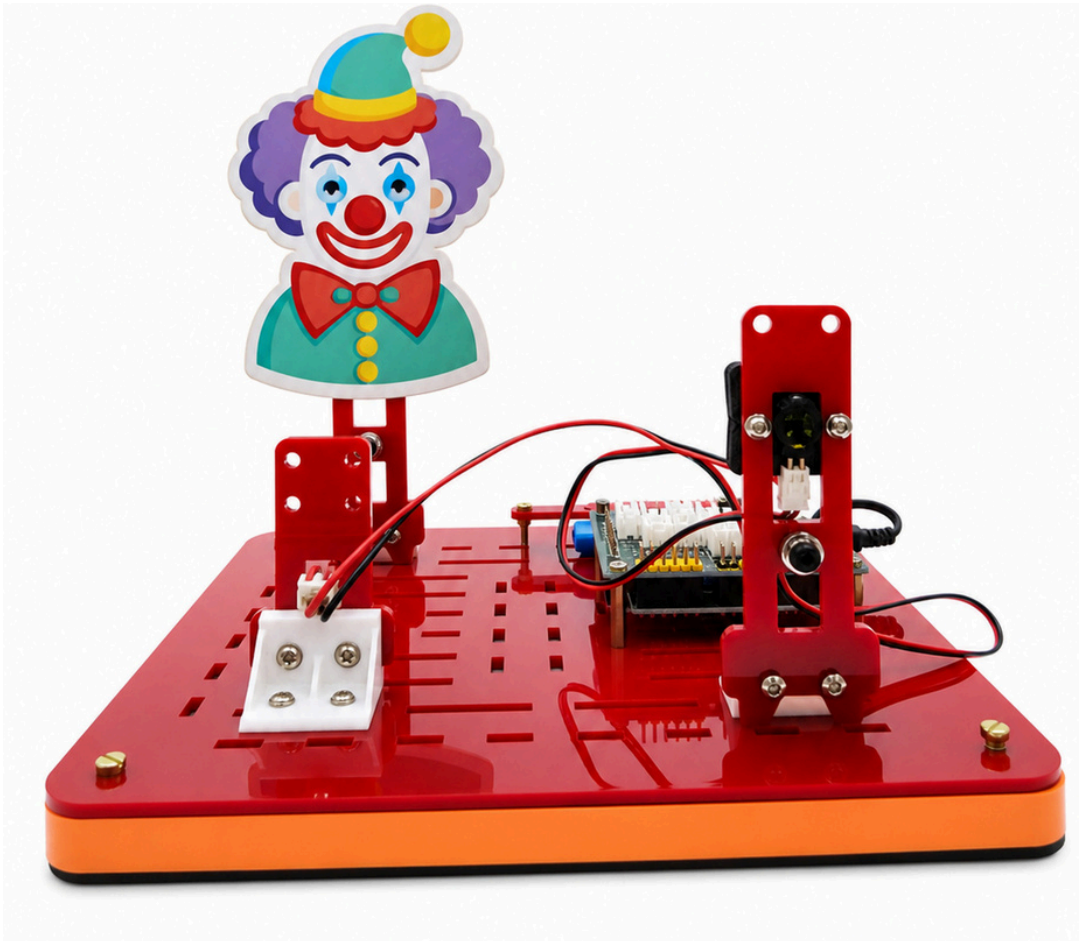
Step 7 – Attach the small stand to the board using an L clamp along with a 3mm nut and bolt.



Step 8- insert rgb led into to the hole in the stand

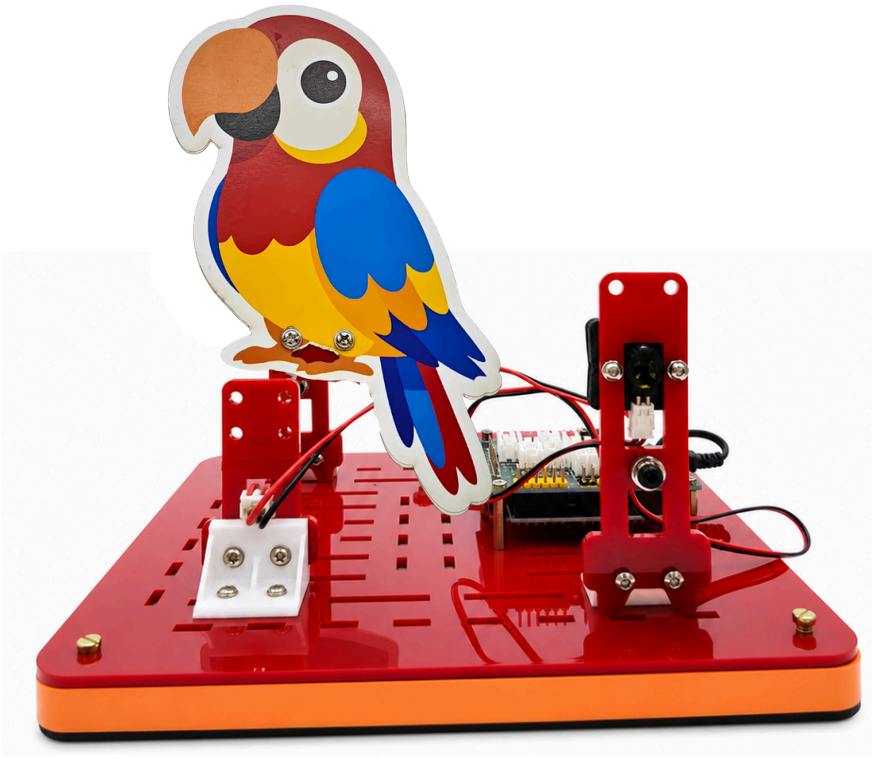


Press the push switch and watch the template change through three different colors: red, green, and blue.



Here just replace the joker template with parrot template





code

```
const int pushSwitch = 5;
const int buzzer = 6;
const int rgbRed = 10;
const int rgbGreen = 12;
const int rgbBlue = 13;

void setup() {
  pinMode(pushSwitch, INPUT_PULLUP);
  pinMode(buzzer, OUTPUT);
  pinMode(11, OUTPUT);

  pinMode(rgbRed, OUTPUT);
  pinMode(rgbGreen, OUTPUT);
  pinMode(rgbBlue, OUTPUT);

  digitalWrite(11, HIGH);  // D11 HIGH forever
}

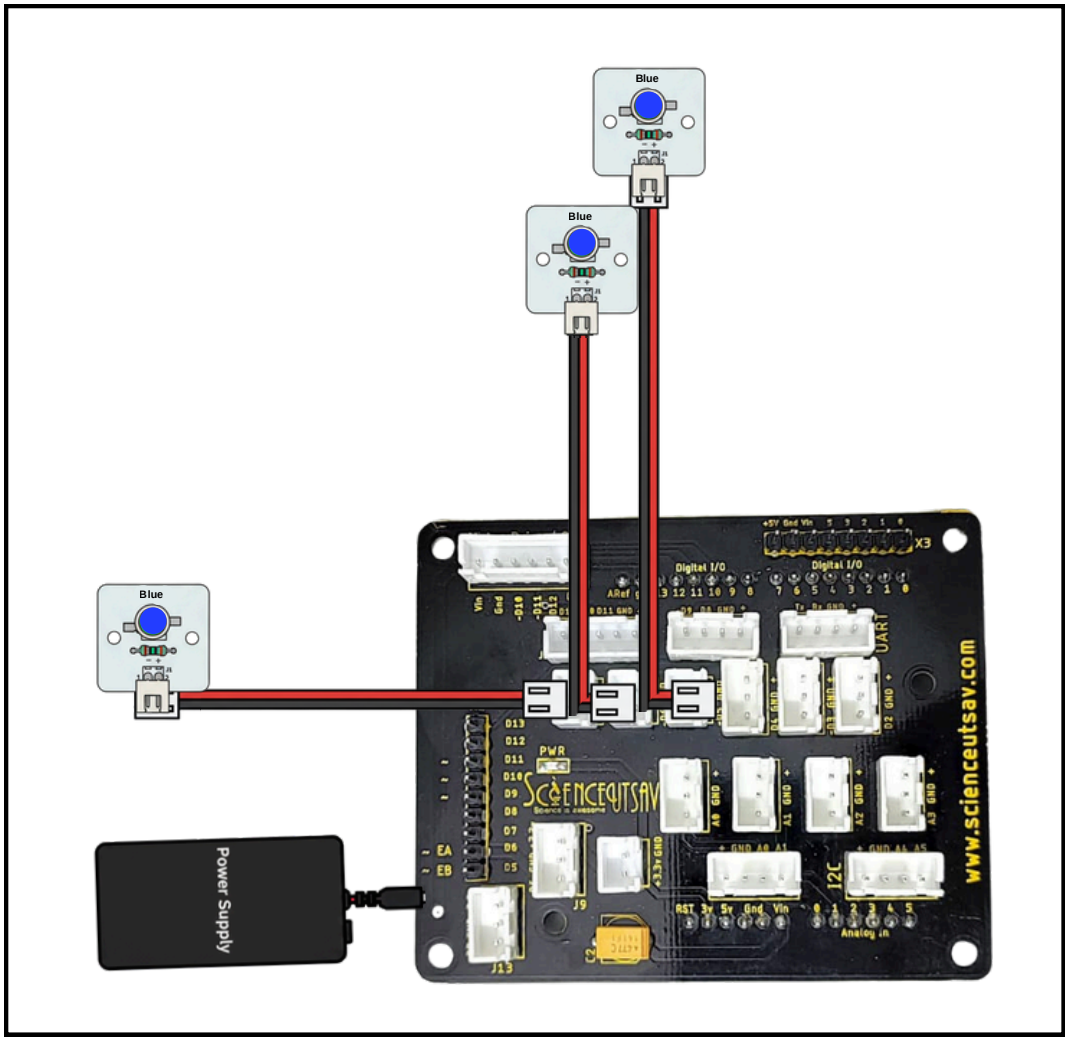
void loop() {
  int switchState = digitalRead(pushSwitch);
  if (switchState == LOW) {
    // Switch pressed
    // Red + buzzer ON
    digitalWrite(rgbRed, HIGH);
    digitalWrite(rgbGreen, LOW);
    digitalWrite(rgbBlue, LOW);
    digitalWrite(buzzer, HIGH);
    delay(500);
}
```

```
// Green + buzzer OFF
digitalWrite(rgbRed, LOW);
digitalWrite(rgbGreen, HIGH);
digitalWrite(rgbBlue, LOW);
digitalWrite(buzzer, LOW);
delay(500);

// Blue + buzzer ON
digitalWrite(rgbRed, LOW);
digitalWrite(rgbGreen, LOW);
digitalWrite(rgbBlue, HIGH);
digitalWrite(buzzer, HIGH);
delay(500);
}
else {
// Switch not pressed
digitalWrite(rgbRed, LOW);
digitalWrite(rgbGreen, LOW);
digitalWrite(rgbBlue, LOW);
digitalWrite(buzzer, LOW);
}
```

EXPERT PROJECT

The circuit diagram of the project is given below

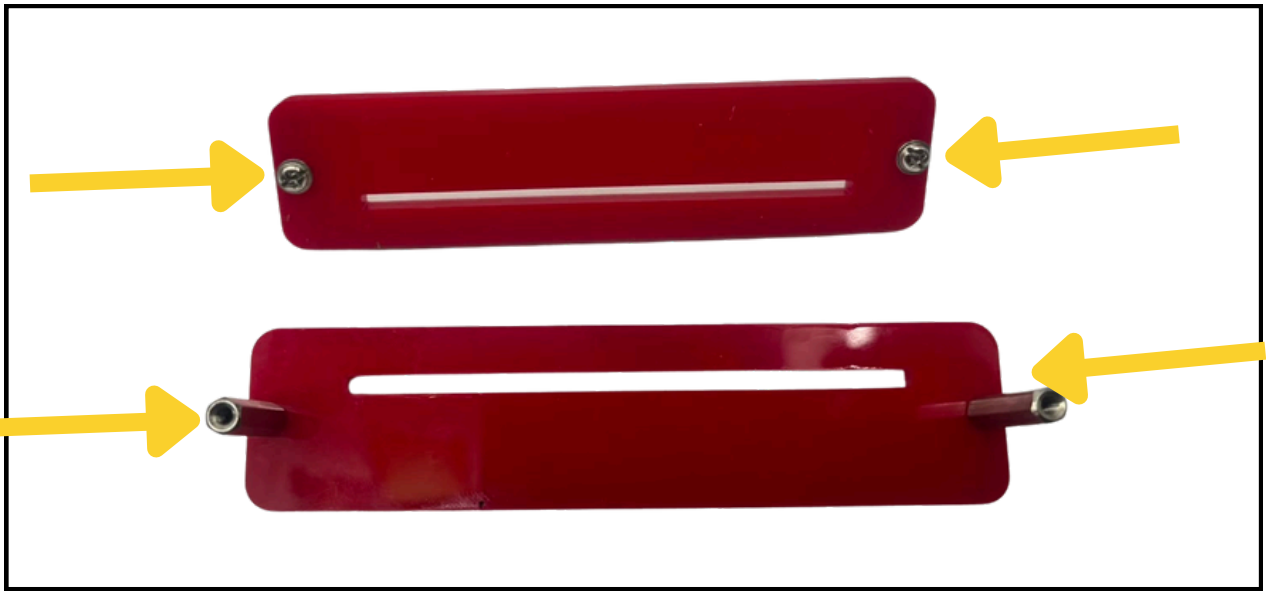


CONNECTION:

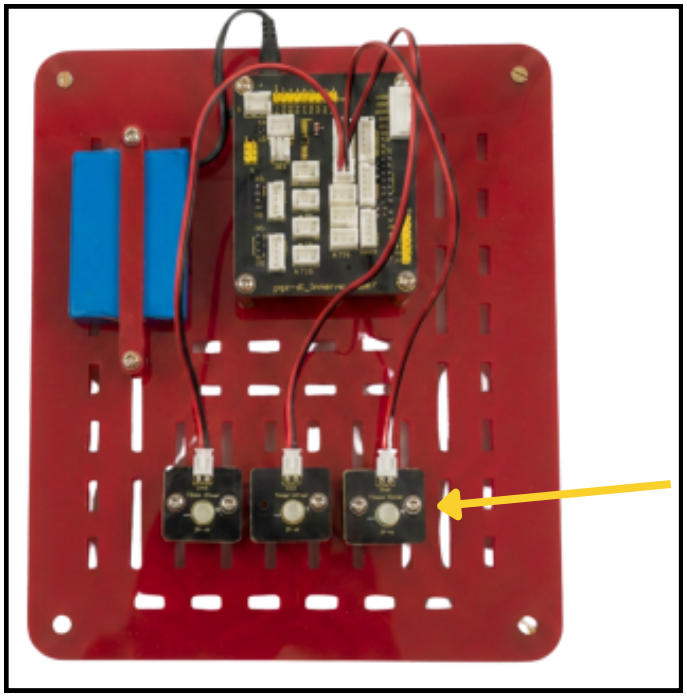
1.) RGB LED IN D10 D11 D12 D13

2.) PUSH SWITCH IN D5

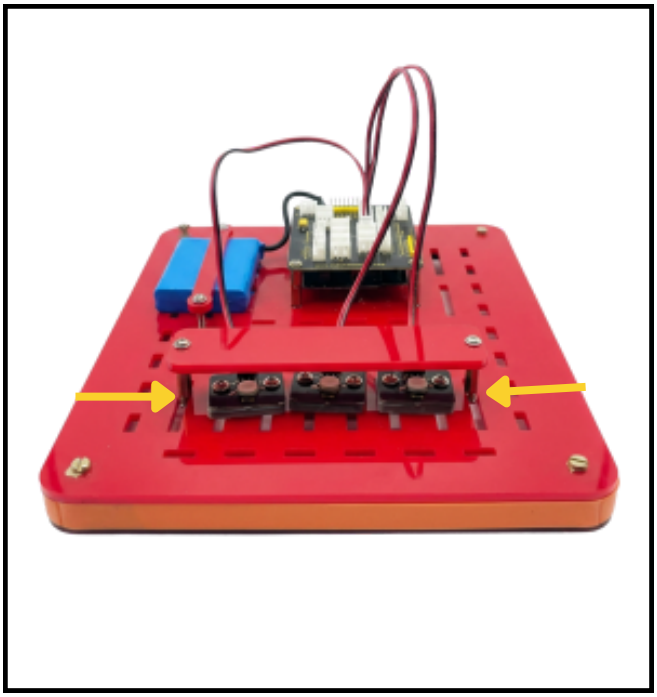
Step 1- Take below shown structure and attach 3mm bolt from side and spacer at the other side of it



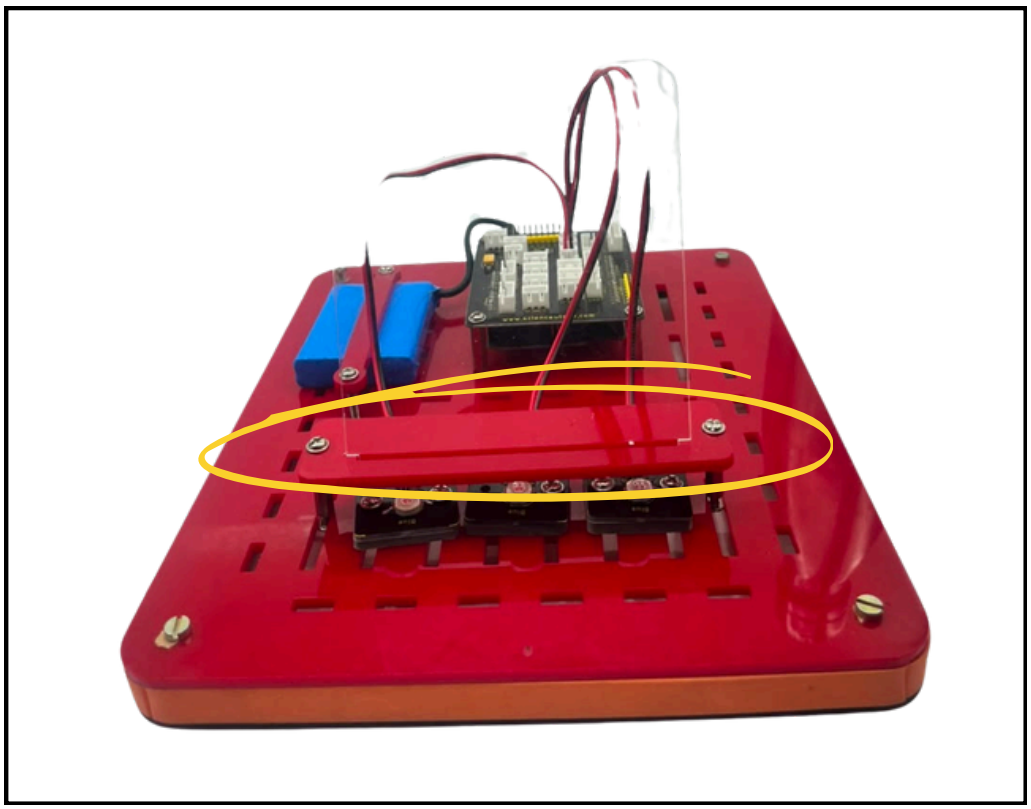
Step 2- Connect the led with D5,D6,D7 and attch the led as shown in the figure with 3mm nut bolt



Step 3- Attach prepared structure on the top of led and add 3mm bolt from the bottom of the board



Step 4- Take transperant acrylic structure and insert on the top of led



code

```
void setup() {
 pinMode(5, OUTPUT); //
pinMode(6, OUTPUT);
pinMode(7, OUTPUT);

}

void loop() {
 digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);

}
```